

SC Honors Geometry - Nappier
Homework 11.3 – Areas of Polygons

Name: _____

Diagonal formulas.

1. A rhombus has diagonals 12 and 16. Find its area.
2. A kite has diagonals 10 and 18. Find its area.

Regular polygons. Exact where marked, else nearest tenth.

3. A regular hexagon has side 8. Find its area EXACTLY.
4. A regular pentagon has side 10. Find its area.
5. An equilateral triangle has side 9. Find its area two ways:
with the apothem formula and with $A = s^2\sqrt{3}/4$.
6. A square has apothem 7. Find its side and its area.

Central angles.

7. Find the central angle of a regular octagon.
8. A regular decagon has side 4. Find its area (nearest tenth).

Working backward.

9. A rhombus has area 84 and one diagonal 8. Find the other diagonal.
10. A regular hexagon has apothem $6\sqrt{3}$. Find its side length and its area (nearest tenth).